

Shanze Owais

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EDUCATION

Michigan State University – East Lansing, MI

May 2026

Bachelor of Science in Data Science – Dean's List

Relevant Coursework: Computational Modeling and Data Analysis, Machine Learning, Matrix Algebra with Computational Applications, Multivariable Calculus, Probability and Statistics

EXPERIENCE

Undergraduate Research Assistant – Michigan State University / East Lansing, MI

August 2024 – Present

Data Collector

- Collaborated on the NIH-funded GOAL study, analyzing data to promote healthy eating and physical activity in adolescents.
- Collected and analyzed data from middle school students using objective measurement techniques, improving research quality and accuracy.
- Trained in data collection methods, including anthropometrics, enhancing research quality and ensuring compliance with ethical standards.

Biwako Sushi – Saline, MI

April 2021 – Present (Seasonal)

Main Waitress

- Provided exceptional customer service in a fast-paced environment, efficiently managing up to 10 tables simultaneously.
- Processed cash transactions accurately, ensuring precise daily sales reporting and contributing to operational efficiency.

PROJECTS

NLP-Based Sentiment Analysis of Tweets

- Developed a sentiment analysis model using NLP techniques, Logistic Regression, and TF-IDF vectorization to classify tweets as Positive, Negative, or Neutral, while preprocessing the dataset and visualizing key sentiment patterns using word clouds.
- Evaluated model performance with key classification metrics including precision, recall, F1-score, and accuracy, achieving an accuracy of 79% on the test set.
- Deployed the model as a Flask API, enabling real-time sentiment prediction for new tweets, allowing integration with external applications for dynamic analysis.

Gemini Clone (React & Vite)

- Developed a clone of the Gemini application using React and Vite, replicating the user interface and core functionality.
- Integrated the Gemini API for real-time data, ensuring smooth interaction and data flow between the frontend and backend.
- Focused on optimizing performance and responsiveness, enhancing the user experience with modern web development practices.

Customer Segmentation Using K-Means Clustering

- Implemented K-Means clustering to segment customer data based on demographics, behavior, and financial metrics, using Python (Pandas, NumPy) and the Elbow Method for optimal clustering.
- Visualized and interpreted results to identify key customer groups, providing actionable insights for targeted marketing and resource optimization.

ACTIVITIES

Software Engineering and Computing Society – East Lansing, MI

June 2024 to Present

- Actively contributed to open-source projects, collaborating with peers to enhance coding skills and software development practices.
- Participated in coding challenges, applying problem-solving skills in real-time scenarios.

Alpha Phi Omega – East Lansing, MI

September 2023 to Present

Leadership and Outreach Chair / Rush Committee

- Led leadership workshops and outreach programs, fostering community engagement and professional development, as well as coordinated rush activities to successfully recruit new members.
- Committed 50+ hours to volunteering initiatives with organizations such as Girl Scouts and the Red Cross.

SKILLS/INTERESTS

Programming Languages: Python, R, SQL, HTML/CSS/JavaScript, C++

Framework/Libraries: Pandas, NumPy, Matplotlib, Seaborn, ggplot2 (Tidyverse), React.js, Next.js, scikit-learn, NLTK

Data Tools: Power BI, Tableau, Excel

Foreign Languages: English, Urdu, Hindi

CERTIFICATIONS

Adult and Pediatric First Aid/CPR/AED – American Red Cross

Credential ID 01LTTP3